

Global Renewable Energy in 2025: How Far Have We Come?

A data-driven overview of renewable energy across primary energy, electricity, and leading technologies

Core question

Renewables are scaling quickly — but fossil fuels still define the global energy system.

75% of global GHG emissions still come from fossil fuels

The fossil-fuel problem is both climate and health

Energy combustion drives three-quarters of greenhouse gases and at least 5 million premature deaths annually from

Global GHG emissions

75%

Premature deaths / year

5M+

Required direction

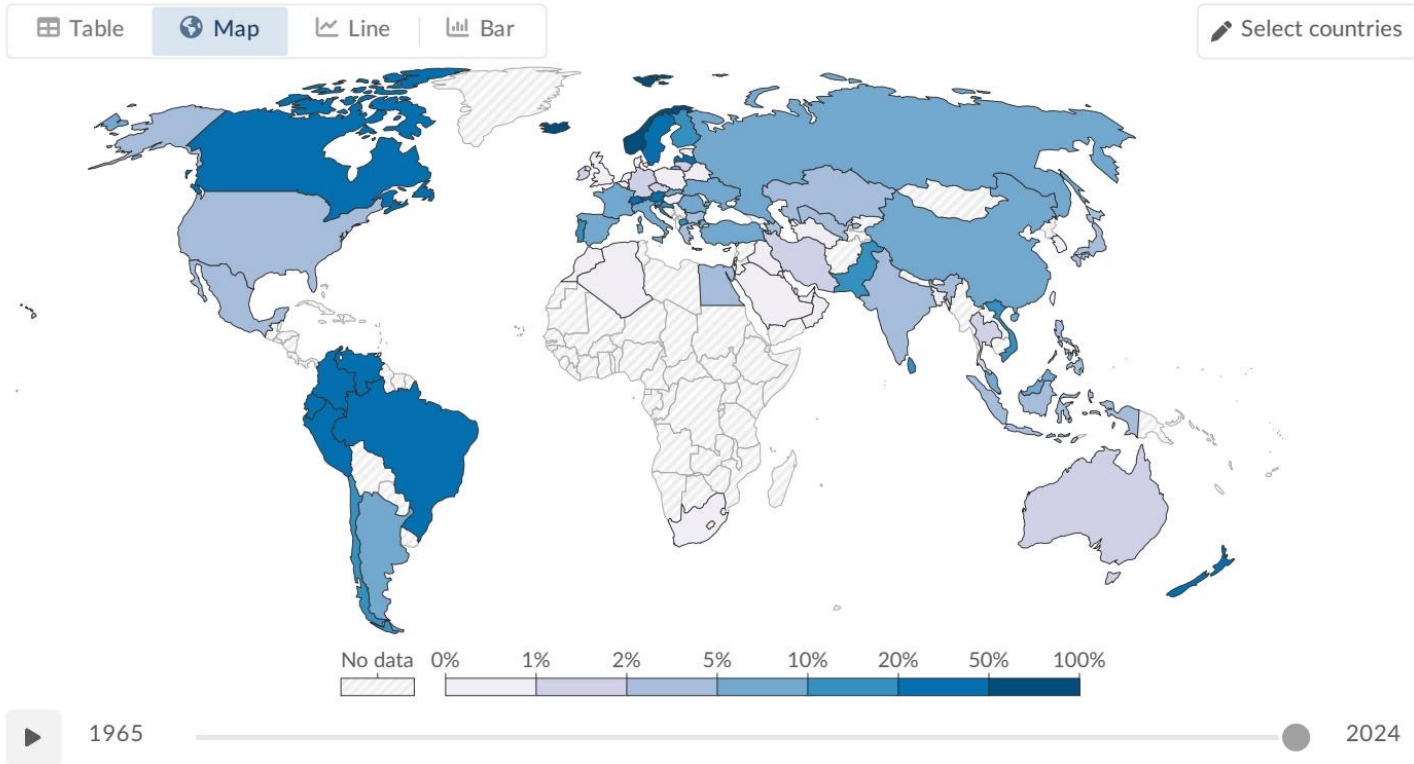
Low-carbon

Since the Industrial Revolution

Fossil-fuel dominated energy systems

Transition imperative: renewables + nuclear

Renewables now supply ~14% of global primary energy



Data source: Energy Institute - Statistical Review of World Energy (2025) – [Learn more about this data](#)
OurWorldinData.org/energy | CC BY

One-seventh

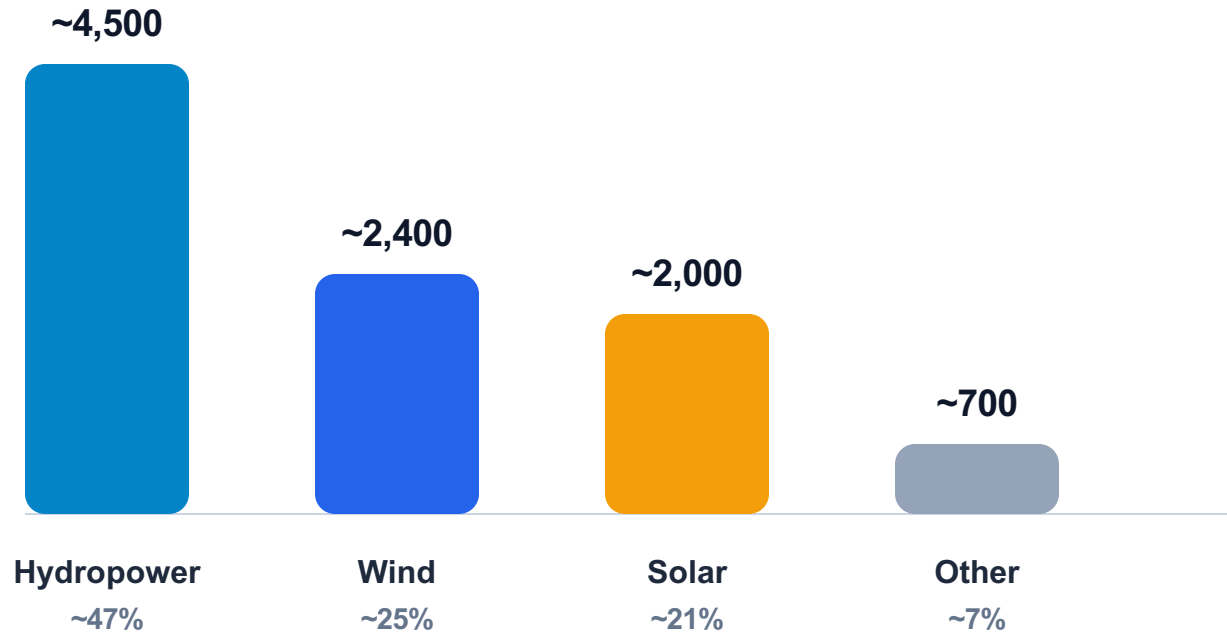
~14%

of global primary energy

- Primary energy includes electricity, transport, and
- Measured using the substitution method.
- Transport and heating keep total energy share be
- Norway, Brazil, and New Zealand stand out; Mido

Renewable electricity generation is led by hydro, with wind and solar closing fast

Approx. global generation, 2024 (TWh)



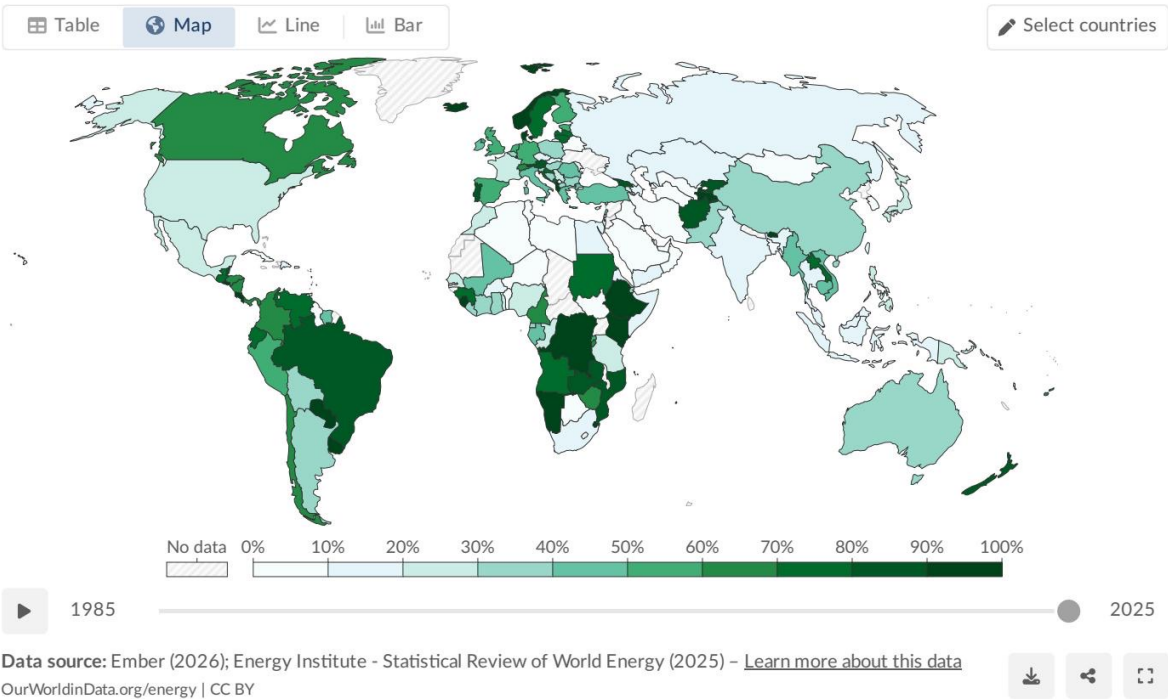
Strategic readout

- Hydropower remains the anchor of renewable power
- Wind + solar now total ~4,400 TWh — nearly hydro
- Acceleration has been strongest since around 20
- Other renewables contribute ~700 TWh, or ~7%.

Almost one-third of global electricity now comes from renewables

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.

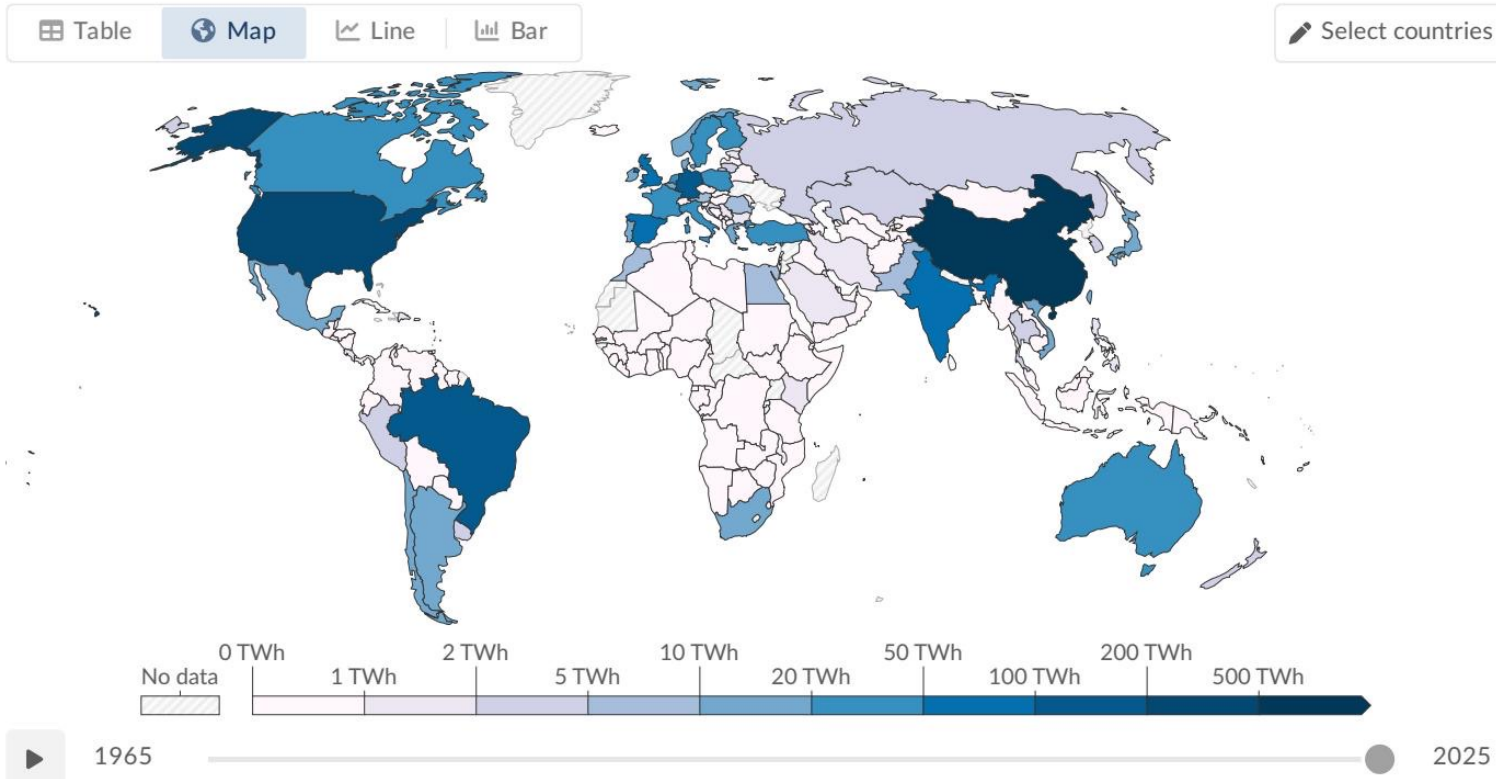


Selected renewable electricity shares

Country / region	Share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%

World average: ~30% — far above primary energy's ~14%.

Hydropower remains the largest renewable source at roughly half of generation



~4,500 TWh

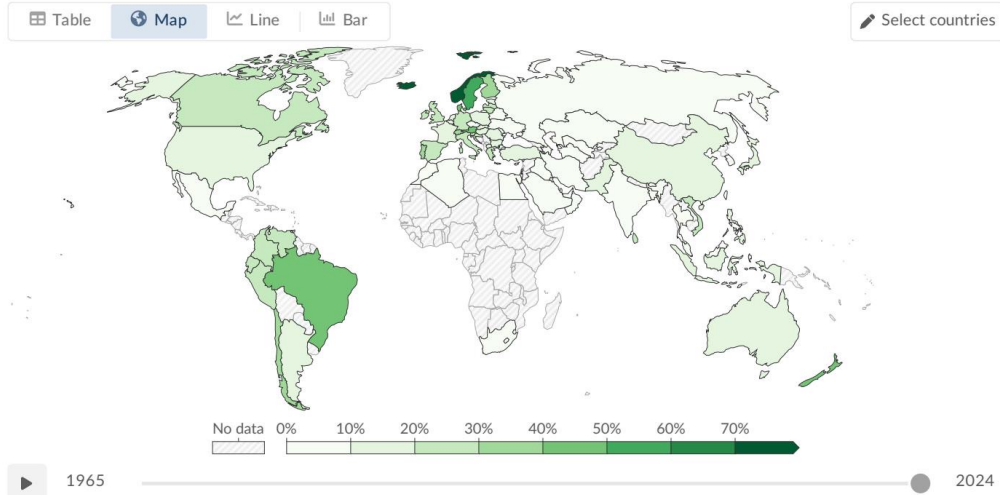
global generation, 2024

~47%

of renewable electricity

- Largest renewable source, operating at scale
- Top producers include China, Brazil, and Canada
- Growth potential is more limited than wind and solar

Wind and solar are the fastest-growing technologies, together nearly matching hydro



Generation and share of renewable electricity

Wind



~2,400 TWh

~25%

Solar



~2,000 TWh

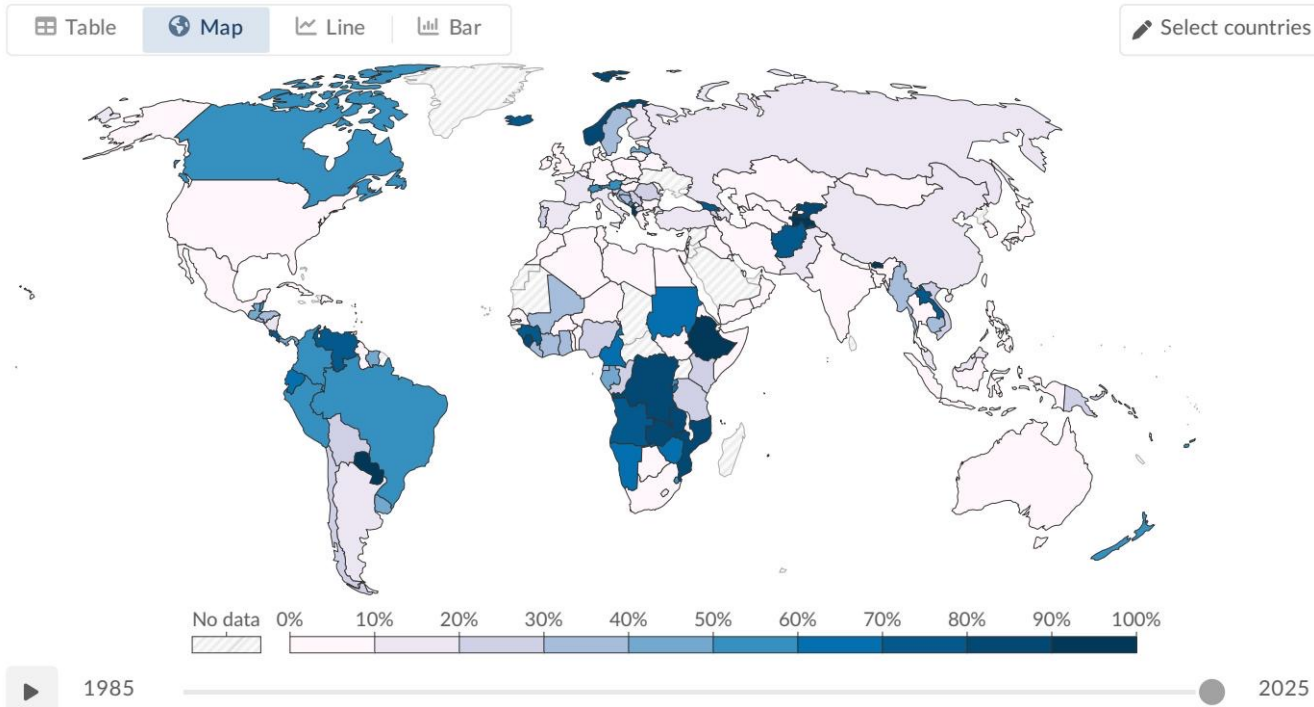
~21%

What matters for strategy

- Solar is the fastest-growing source, with dramatic growth since 2010 driven by falling costs and policy support, especially in Europe, China, and the U.S.

Wind + solar: ~4,400 TWh, almost equal to hydropower's ~4,500 TWh.

Regional gaps remain stark: hydropower-rich regions lead, fossil regions lag



Uneven transition patterns

- Scandinavia, Brazil, and Canada benefit from abundant hydropower
- Denmark and Germany have invested heavily in wind and solar
- Many African and Middle Eastern countries remain below 10%
- Middle Eastern nations remain almost entirely fossil-fuel dependent

Infrastructure and investment gaps define where acceleration is needed

Key takeaways: renewables are growing fast, but the transition must accelerate

~14%

of primary energy is renewable

~30%

of electricity is renewable

~47%

of renewable electricity still comes from hydro

~4,400 TWh

from wind + solar combined

<10%

in many African and Middle Eastern countries

Forward-looking call

Electricity is moving first; policy, capital, grids, and end-use electrification must now accelerate transport and heat